Introduction to Artificial Intelligence (Fall 2026)
Homework Project

LLM-A*: Large Language Model Enhanced Incremental Heuristic Search for Path Planning

This assignment is based on the paper *LLM-A*: Large Language Model Enhanced Incremental Heuristic Search*. The objective of this project is to understand how large language models (LLMs) can be combined with the A* search algorithm to improve path-planning efficiency.

- Students must first understand the algorithm presented in the paper and reproduce the main experimental results reported in Table 1.
- Students must then analyze the limitations of the original method and develop their own improved algorithm.
- Students may use any accessible LLM to reproduce Table 1 and develop their proposed method.
- The submitted code will be evaluated primarily using **GPT-4o mini** as the designated backbone model.
- Additional LLMs may also be used to evaluate the robustness and generalizability of the proposed method.

[Team Formation]

- This project may be completed either individually or as a team.
- Forming a team is optional.
- A team may consist of up to **three students**.
- Regardless of whether the project is completed individually or as a team, each student is responsible for accurately reporting their own contributions.

[Relevant Links]

- Paper Project Page
  https://silinmeng0510.github.io/llm-astar/
- Paper
  https://arxiv.org/pdf/2407.02511
- Original GitHub Repository
  https://github.com/SilinMeng0510/llm-astar

[Project Schedule]

The project will proceed according to the following schedule:

1. **Project Period: September 21 - November 22, 23:59**
    - Students must complete and submit their experiments, project report, and code during this period.
2. **Peer Review Period: November 23 - December 6, 23:59**
    - Each student or team must review the project results of another team.
    - The assignment of teams for peer review will be announced separately.
3. **Response to Peer Review Period: December 7 - December 16, 23:59**
    - After receiving a peer review from another team, each student or team must evaluate the review and submit a response.
    - The response must explain whether the reviewers' comments are accurate and helpful and provide the reasoning behind this assessment.

All materials must be submitted within their respective submission periods.

[Evaluation Environment and Policy]

- The assignment will be evaluated based on both the performance of the submitted method and the quality of the report. Performance is an important consideration, but it is not the only evaluation criterion.
- Performance will be evaluated separately by the teaching assistants using hidden environments and datasets, which will not be disclosed in advance.
- GPT-4o mini will be used as the primary backbone model for evaluation. Additional models may also be used to examine the robustness and generalizability of the submitted methods.
- Results obtained only from the publicly available environments should not be regarded as the final evaluation results.
- The submitted code must follow the required input and output formats and run on previously unseen environments without manual modification. Methods tuned specifically to the public environments (e.g., hard-coding for specific maps) will be penalized.
- The submitted code and the report must be consistent. The method described in the report must be the one implemented in the submitted code.

[Report Evaluation Criteria]

- Reports that satisfy all basic requirements will start from a baseline score. Points will be deducted for missing or insufficient content and added for particularly strong work.
- The report is evaluated on how clearly and convincingly the work is explained and supported, not on the performance numbers themselves.

The report will be evaluated on the following aspects:

- **Completeness**: All required sections are included: experimental setup, methodology, Table 1 reproduction results, proposed algorithm, experimental results, analysis, and limitations. The report follows the ACL template and includes proper references.
- **Reproduction**: Table 1 is reproduced, the results are compared with those reported in the paper, and any discrepancies are explained. Results that differ from the paper will not be penalized if the differences are properly analyzed.
- **Methodology**: The proposed method is clearly described, its differences from the original LLM-A* are explicit, and the design is motivated by limitations identified in the original method.
- **Experiments and Analysis**: The proposed method is fairly compared with A* and LLM-A*. Results are interpreted rather than merely listed, including reasons for improvements, failure cases, and ablation or step-by-step comparisons where appropriate.
- **Limitations**: The limitations of the proposed method are honestly discussed.
- **Reproducibility**: The report provides sufficient information (LLM settings, prompts, environment configuration, and execution instructions) for another person to reproduce the experiments.

Additional credit may be given for:

- Evaluation of generalization using other LLMs or new environments
- Clear presentation using pseudocode, equations, or figures
- Well-explained failed experiments and cost-efficiency analysis

Serious issues that may result in a score cap or major deduction:

- Missing Table 1 reproduction or missing experimental results
- An unclear or incomprehensible proposed method
- Inconsistencies between the report and the submitted code
- Unsupported or exaggerated claims
- Plagiarism or fabrication of results, which will result in a score of zero or be referred for separate review

[Academic Integrity and AI Usage Policy]

- The use of AI tools (e.g., ChatGPT, Claude) to assist with writing the report is permitted. However, all AI-generated text must be carefully reviewed and revised in your own words, and you are fully responsible for the accuracy of everything in your report.
- The report must reflect your own work. Descriptions of methods, experiments, and results must match what you actually implemented and conducted.
- Submitting AI-generated text without meaningful revision, or including content that does not reflect your actual work (e.g., fabricated results, nonexistent references, or descriptions of experiments that were not conducted), will be considered a violation of academic integrity and may result in penalties, including a score of zero.
- Copying is strictly forbidden. This includes copying reports or code from other teams, previous submissions, or online sources without proper attribution. Any identified case will result in a score of zero and may be referred for further action according to the course policy.
- Using the official LLM-A* code for reproduction and building on it is allowed, but all external code and resources must be clearly cited in the report.

[Peer Review]

Peer review is a required part of this project.

- Each student or team must review the project results of another team.
- The review must provide specific, constructive, and evidence-based feedback on the other team's methodology, experimental setup, results, analysis, limitations, and reproducibility.
- A peer-review submission form will be provided separately.
- The assignment of teams for peer review will be announced separately.

The peer review must include the following:

1. **Three Strengths**
2. **Three Weaknesses**
3. **Three Suggestions**

Each individual strength, weakness, and suggestion must be explained in at least **two to three sentences**. Accordingly, the review must contain a total of nine items, and each item must be explained in sufficient detail.

Reviewers must not provide only a brief conclusion such as "good" or "insufficient." Each assessment must include a clear explanation and specific evidence supporting the reviewer's judgment.

After receiving a peer review, each student or team must review it and submit a response. The response must include:

- Whether the student or team agrees or disagrees with each comment
- The reasoning behind their judgment

[Submission Materials]

Please submit the following materials:

1. Project Report

File name: TeamName_Report

- The report must clearly describe the experimental setup, methodology, Table 1 reproduction results, proposed algorithm, experimental results, and analysis.
- The report must describe not only the strengths of the proposed method but also its limitations.
- The report must provide sufficient information for another person to reproduce the experiments.

2. Code Files

File name: **TeamName_CodeFile.zip**

1. All code files must be compressed and submitted as a single **.zip** file.
2. The **.zip** file must include a **requirements.txt** file listing all packages required for installation.
3. The submitted code must include a **README** file that clearly explains how to reproduce the Table 1 experiments and run the proposed algorithm.
4. The code must support the input and output formats specified in the baseline repository so that it can be evaluated on hidden maps.
5. The code must run on hidden environments without requiring manual modification of the algorithm or data format.
6. API keys, access tokens, passwords, and other credentials must not be included in the submitted files.

3. Team Contribution Report (Individual Submission)

1. Every student must submit an individual contribution report.
2. The contribution report must be at least **half a page long**.
3. Students who complete the project as a team must provide a specific and factual description of the tasks performed by themselves and each teammate, the deliverables produced, the approximate dates of the work, and the extent of each member's contribution.
4. Students who complete the project individually must describe their overall work process, tasks performed, deliverables produced, and approximate dates of the work according to the same length requirement.
5. This is an individual submission. Students must not discuss or share their contribution reports with their teammates.

6. The report must include specific and factual details, such as modules implemented, report sections written, experiments conducted, issues or pull requests handled, and meetings attended.
7. We will check consistency across team members' reports and compare them with the project artifacts to identify discrepancies and potential free-riding.
8. Failure to submit the report, or submission of a vague or insufficient report, may result in a score of zero for the project participation component.
9. If serious teamwork issues are identified, an individual student's project score and, if necessary, the team's score may be adjusted according to the course policy.

4. Peer Review of Another Team

- Each student or team must submit a review of the assigned team's project results during the peer review period.
- The review must follow the provided peer-review submission form and include three strengths, three weaknesses, and three suggestions.
- Each strength, weakness, and suggestion must be explained in at least two to three sentences.

5. Response to Peer Review

- Each student or team must review and respond to the comments received from another team during the response period.
- The response must state whether the student or team agrees or disagrees with the review comments and explain the reasoning behind their judgment.

Please ensure that all required materials are included and submitted within the designated periods.

[Academic Integrity]

- All project reports must be written independently by the submitting student or team.
- Copying another student's or team's report, experimental results, analysis, or written content is strictly prohibited.
- Any external resources, source code, or ideas used in the project must be appropriately acknowledged and cited.
- If substantial similarities are identified between submissions or copying is suspected, additional verification may be required. This may include requests for code revision history, experimental records, or an oral explanation of the submitted work.
- Confirmed violations will be handled according to the course's academic integrity policy.

[TA Office Hours]

- **Weeks:** Week 6 and Week 10 (Wednesday)
- **Location:** Office 27303, Engineering Building 2
- **Time:** 5 PM - 6 PM

[Communication]

All communication regarding individual questions and assignment-related matters will be handled **by email only**.

Email Information

- **Contact Email: swe3011_g@g.skku.edu**
- **Suggested Email Subject Format: [LLM-A* Assignment] Team name**
- When sending questions, please include your execution environment, the LLM and model version used, relevant error messages, and what you have already tried.